```
import re
text = "cat bat mat"
result = re.findall("at", text)   # findall() returns all matches
print("All matches:", result)
```

```
All matches: ['at', 'at', 'at']
```

```
import re
text = "I love Python"          # search() finds occurrence anywhere
print(re.search("Python", text))
```

```
<re.Match object; span=(7, 13), match='Python'>
```

```
import re
text = "i am learning python"           # match() checks only at the beginning
print(re.match("python", text))
```

```
None
```

```
import re
text = "I like Python"            # '$' checks if string ends with pattern
print("match():", re.match("Python", text))
print("search():", re.search("Python", text))
```

```
match(): None
search(): <re.Match object; span=(7, 13), match='Python'>
```

```
import re
text = "Room number is 101"             # '\d' finds individual digits
result = re.findall(r"\d", text)
print("Digits:", result)
```

```
Digits: ['1', '0', '1']
```

```
import re
text = "Marks are 85 and 100"                    # '\d+' finds complete numbers
result = re.findall(r"\d+", text)
print("Numbers:", result)
```

```
Numbers: ['85', '100']
```

```
import re
text = "Python123"                     # '\w' matches letters, digits, underscore
result = re.findall(r"\w", text)
print("Characters:", result)
```

```
Characters: ['P', 'y', 't', 'h', 'o', 'n', '1', '2', '3']
```

```
import re
text = "Python is easy"                  # '\w+' matches full words
result = re.findall(r"\w+", text)
print("Words:", result)
```

```
Words: ['Python', 'is', 'easy']
```

```
import re
text = "A B C"                                  # '\s' finds spaces
result = re.findall(r"\s", text)
print("Spaces:", result)
```

```
Spaces: [' ', ' ']
```

```
s = input("Enter a string: ")
digits = ""

for ch in s:
    if ch.isdigit():
        digits += ch

print("Digits:", digits)
```

```
Enter a string: gvq7jgv6jqhg7
Digits: 767
```

```
import re

s = input("Enter a sentence: ")

numbers = re.findall(r'\d+', s)

print("Numbers:", numbers)
```

```
Enter a sentence: j7ruh6uh6wuh44t4
Numbers: ['7', '6', '6', '44', '4']
```

```
s = input("Enter a string: ")

words = s.split()

print("Words:", words)
```

```
Enter a string: hgb itnh urh fh h
Words: ['hgb', 'itnh', 'urh', 'fh', 'h']
```

```
import re

text = input("Enter a string: ")
spaces = len(re.findall(" ", text))

print("Number of spaces:", spaces)
```

```
Enter a string: df g dng  f fhnfgn n ryjn ryn
Number of spaces: 9
```

```
import re

text = input("Enter a string: ")                   # Regex to find words starting with a capital letter
words = re.findall(r'\b[A-Z][a-z]*\b', text)

print("Words starting with a capital letter:", words)
```

```
Enter a string: uvu Fuvfvdf FU FDY dfvduyfFFF hsvuhb idfAjb
Words starting with a capital letter: ['Fuvfvdf']
```

```
import re

text = input("Enter a string: ")
result = re.sub(r'\d', '#', text)

print(result)
```

```
Enter a string: jh6kjd6gv6dgv6uv6
jh#kjd#gv#dgv#uv#
```

```
import re

number = input("Enter mobile number: ")

if re.fullmatch(r'\d{10}', number):
    print("Valid mobile number")
else:
    print("Invalid mobile number")
```

```
Enter mobile number: 6567676767
Valid mobile number
```

```
import re

email = input("Enter email: ")

if re.match(r'.+@.+\..+', email):
    print("Valid email")
else:
    print("Invalid email")
```

```
Enter email: vihruth999@gmail.com
Valid email
```

```
import re

text = input("Enter a string: ")
result = re.sub(r'\s+', ' ', text)

print(result)
```

```
Enter a string: rthrt     rthrn     hryyj yntny   nr
rthrt rthrn hryyj yntny nr
```

```
import re

text = input("Enter a string: ")

numbers = re.findall(r'\d+', text)

print(numbers)
```

```
Enter a string: iruhb7ng87tgb7tbtt745t735th45t45h745by   87y754utng 45yg75 78
['7', '87', '7', '745', '735', '45', '45', '745', '87', '754', '45', '75', '78']
```